



Network analysis of stress and well-being among college students: A longitudinal study

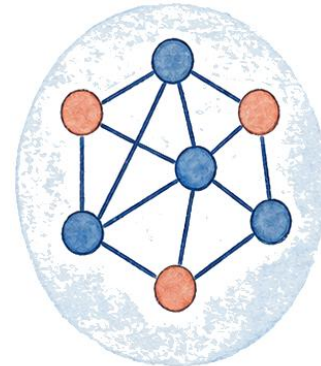
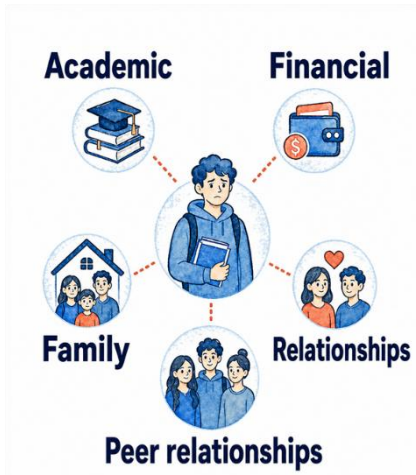


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Objective



Stress & Well-being Network



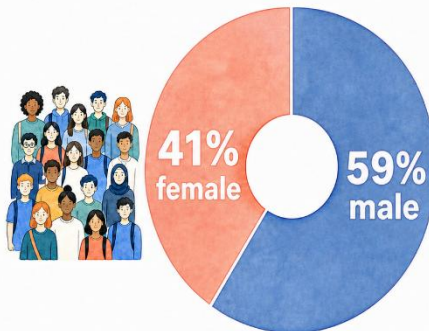
How are stress and well-being connected?



How do these relationships change over time?

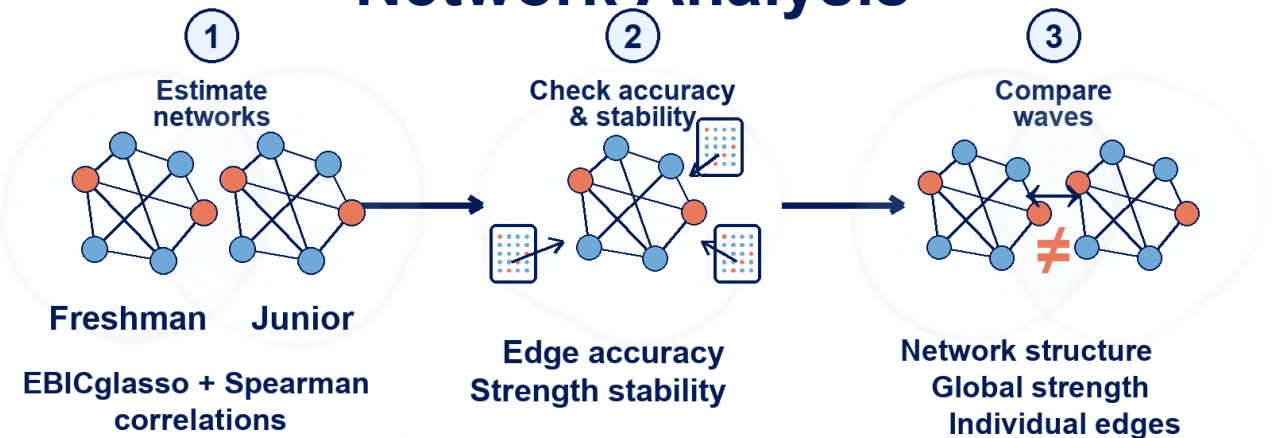
Methods

Panel Data (N = 347)



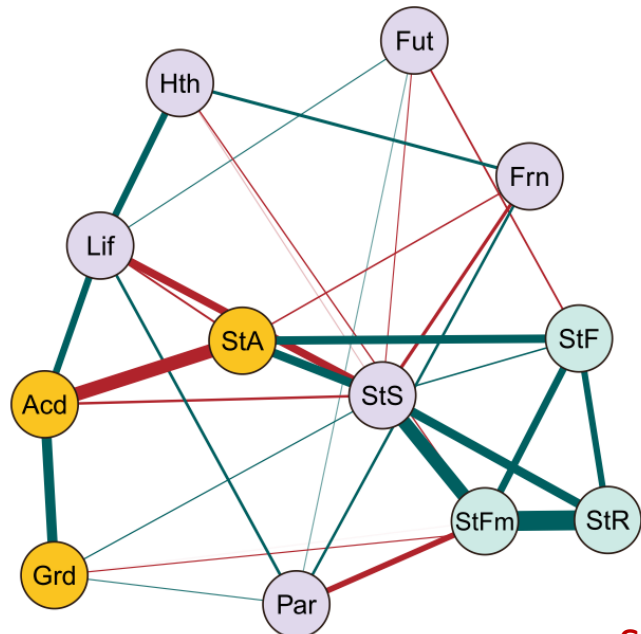
Freshman (Time 1) → Junior (Time 2)
Same students followed over time

Network Analysis

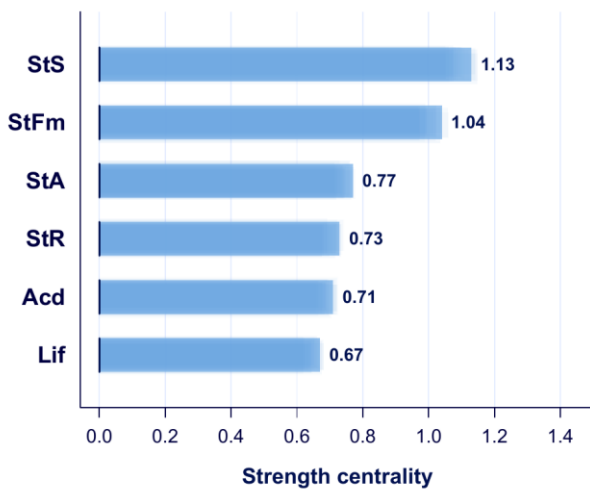


Results

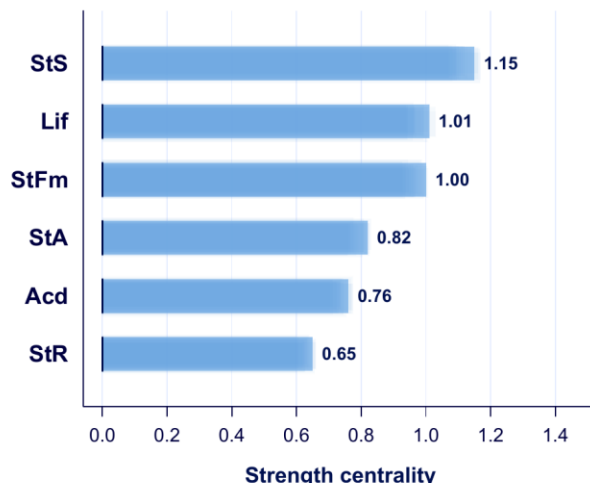
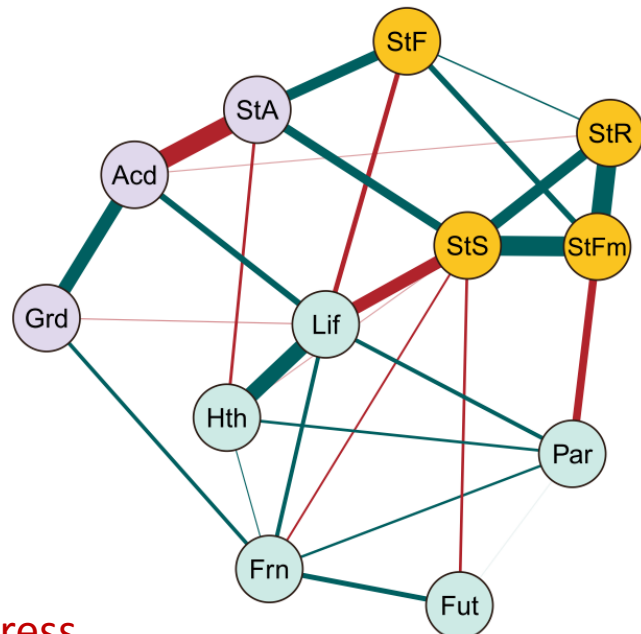
Freshman Year



StS=social stress



Junior Year



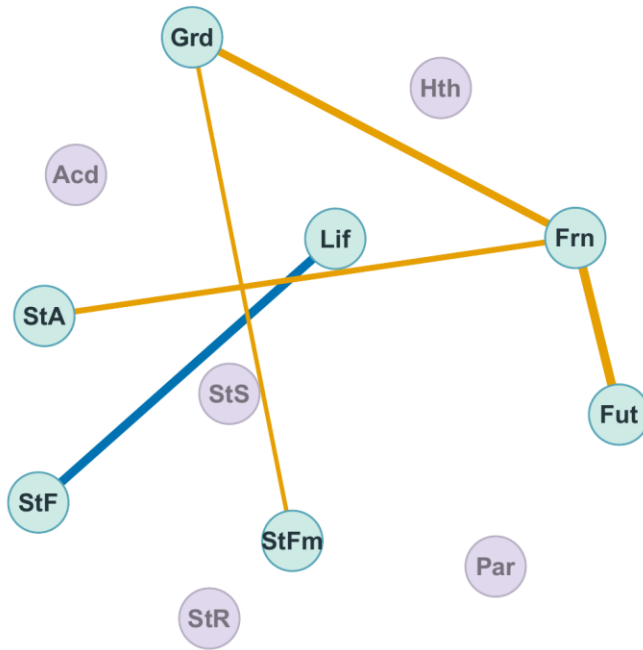
- Social stress was the most central node among the variables in both freshman and junior years.

Guide to Network Nodes



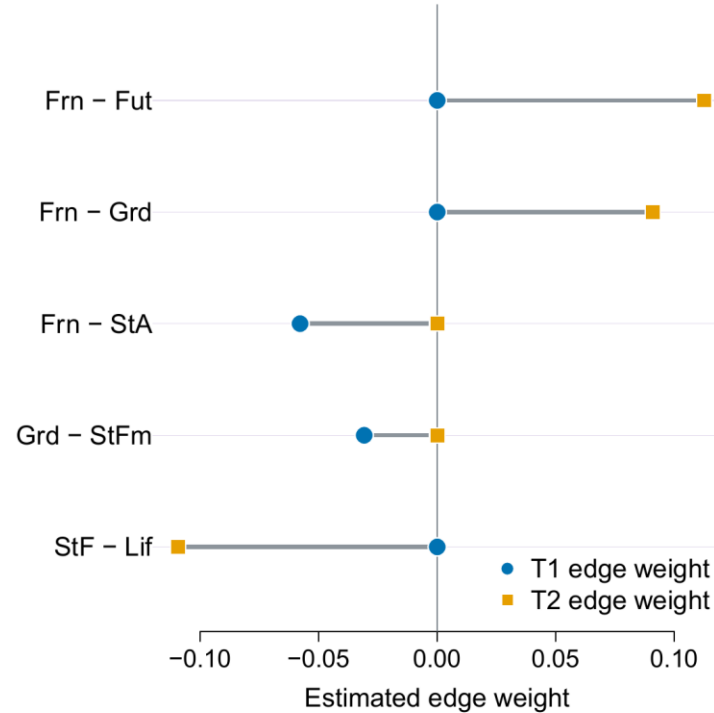
Comparison of junior and freshman years

A. Network location of significant differences



Only NCT edge-invariance tests with unadjusted $p < .05$ are shown; $\Delta = T2 - T1$.

B. T1 and T2 edge weights



- Overall network structure remained stable.
- Close friendships had stronger links with grades and future outlook at T2.
- Negative Frn–StA and Grd–StFm associations were stronger at T1
- Financial stress became more negatively associated with life satisfaction at T2.

Discussion and Conclusion

- Stable overall network, but the roles of close friendships and financial stress may shift as students progress through college.
- These results support sustained peer-support programs and increased financial guidance.
- Edge-level findings are exploratory and based on unadjusted p-values.

Guide to Network Nodes

Stress-related nodes

Higher scores indicate greater perceived stress.

- **StA** — Academic-performance stress
- **StF** — Financial stress
- **StFm** — Parent–adolescent relationship stress
- **StR** — Intimate-relationship stress
- **StS** — Peer-relationship stress

Academic, relationship, and well-being nodes

- **Acd** — Concern about academic performance (after reverse coding)
1 = most bothered; 5 = least bothered
- **Grd** — School grades
1 = low; 5 = high
- **Par** — Closeness with parents
Higher scores = closer relationship
- **Frn** — Having close friends
1 = no; 2 = not really; 3 = yes
- **Hth** — Perceived health
1 = poor; 5 = good
- **Fut** — Future planning
1 = low; 4 = high
- **Lif** — Feelings about life (life satisfaction)
Higher scores = happier and less distressed